3C 1 SHAPES OF MOLECULES AND IONS

SPECIFICATION REFERENCE: 3.16 | 3.17 PART | 3.18 | 3.19

LEARNING OBJECTIVES

- Understand the principles of the electron-pair repulsion theory, used to interpret and predict the shapes of simple molecules and ions.
- Understand the term bond angle.
- Know and be able to explain the shapes of, and bond angles in, $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4.
- Be able to apply the electron-pair repulsion theory to predict the shapes of, and bond angles in, molecules and ions analogous to those mentioned above.

ELECTRON PAIR REPULSION THEORY

The **electron pair repulsion (EPR) theory** states that:

- the shape of a molecule or ion is caused by repulsion between the pairs of electrons, both bond pairs and lone (non-bonding) pairs, that surround the central atom
- the electron pairs arrange themselves around the central atom so that the repulsion between them is at a minimum
- lone pair–lone pair repulsion > lone pair–bond pair repulsion > bond pair–bond pair repulsion.

LEARNING TIP

This theory is sometimes also called the valence shell electron pair repulsion theory, abbreviated to VSEPR.

The first two rules are used to obtain the basic shape of the molecule or ion. The third rule is used to estimate values for the bond angles.

THE SHAPES OF MOLECULES AND IONS

To obtain the shape of a molecule or ion it is first necessary to obtain the number of bond pairs and lone pairs of electrons around the central atom.

The easiest way to do this is by drawing a dot-and-cross diagram. You can then apply the guidelines listed in **table A**.

MOLECULES WITH MULTIPLE BONDS

To determine the shape of a molecule containing one or more multiple bonds, treat each multiple bond as if it contained only one pair of electrons.

EXAMPLE 1. CARBON DIOXIDE, CO_2

The displayed formula for carbon dioxide is O=C=O. There are no lone pairs on the carbon atom.

If each double bond is treated as an electron pair, then the molecule is linear, like $BeCl_2$.

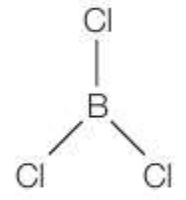
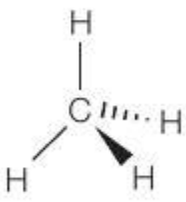
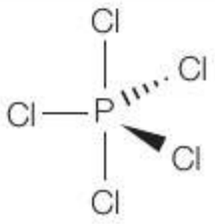
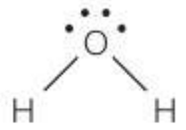
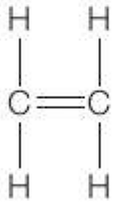
NUMBER OF BOND PAIRS	NUMBER OF LONE PAIRS	SHAPE	EXAMPLE
2	0	linear	
3	0	trigonal planar	
4	0	tetrahedral	
5	0	trigonal bipyramidal	
6	0	octahedral	
3	1	trigonal pyramidal	
2	2	V-shaped	

table A Shapes of molecules.

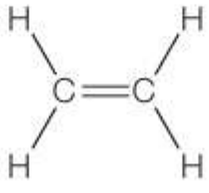
EXAMPLE 2. ETHENE, C_2H_4

The displayed formula of ethene is:



There are no lone pairs on either carbon atom.

Treating each double bond as an electron pair produces a planar molecule with 120° bond angles.



THE BOND ANGLES IN MOLECULES AND IONS

Table B shows the bond angles of a range of molecules and ions.

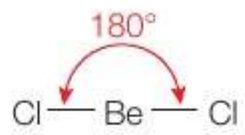
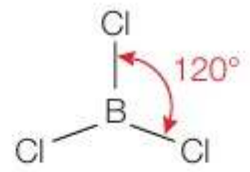
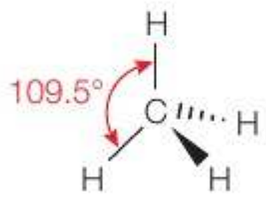
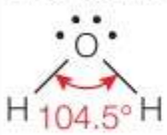
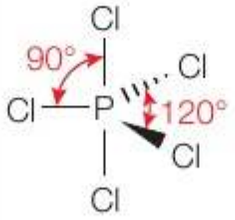
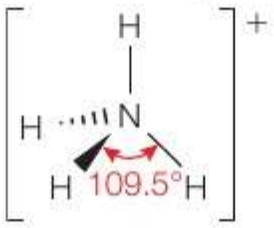
Linear, e.g. $BeCl_2$ The bond angle is 180°. 	Trigonal planar, e.g. BCl_3 The bond angle is 120°. 
Tetrahedral, e.g. CH_4 The bond angle is 109.5°. 	Trigonal pyramidal, e.g. NH_3 The bond angle is 107°.  Lone pair–bond pair repulsion is greater than bond pair–bond pair repulsion, so the angle is slightly less than 109.5°.
V-shaped, e.g. H_2O The bond angle is 104.5°.  Lone pair–lone pair repulsion is greater than lone pair–bond pair repulsion, so the bond angle is even further depressed from 109.5°, and is slightly less than the 107° in NH_3.	Trigonal bipyramidal, e.g. PCl_5 There are two bond angles: 90° and 120°. 
Octahedral, e.g. SF_6 There are two bond angles: 90° and 180°.  The angle between the bonds of two fluorine atoms opposite one another is 180°.	Tetrahedral, e.g. NH_4^+ As with CH_4, the bond angles are 109.5°.  Note the change from 107° in ammonia to 109.5° in the ammonium ion.

table B The bond angles of a range of molecules and ions.

CHECKPOINT

1. (a) Draw a diagram to show the shape of each of the following molecules:

(i) H_2S (ii) PH_3 (iii) PF_3 (iv) SCl_2 (v) AsF_5 (vi) HCN (vii) SO_2

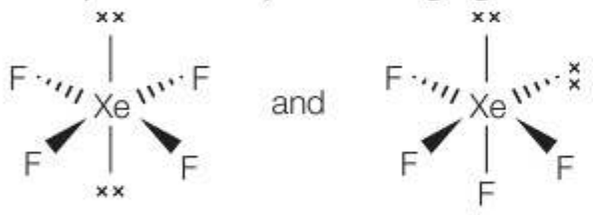
(b) Give the name of each shape.

2. Solid phosphorus pentachloride has the formula $[PCl_4]^+(PCl_6]^-$.

(a) Draw a diagram to show the shape of each ion.

(b) State the bond angles present in each ion.

3. Two possible ways of arranging the bonding pairs and lone pairs of electrons in a molecule of XeF_4 are:



Suggest which of these two arrangements is the more likely and justify your answer.

SUBJECT VOCABULARY

electron pair repulsion (EPR) theory the electron pairs on the central atom of a molecule or ion arrange themselves in order to create the minimum repulsion between them; lone pair-lone pair repulsion is greater than lone pair-bond repulsion, which in turn is greater than bond pair-bond pair repulsion

3C 2 NON-POLAR AND POLAR MOLECULES

SPECIFICATION REFERENCE 3.15 PART

LEARNING OBJECTIVES

- Understand the difference between non-polar and polar molecules and be able to predict whether or not a given molecule is likely to be polar.

SHAPE AND POLARITY

The drift of bonded electrons towards the more electronegative element (see **Topic 3B.2**) results in a separation of charge. This separation of charge is called a **dipole**.

Each of the bonds in a molecule has its own dipole associated with it. The overall dipole of a molecule depends on its shape. Depending on the relative angles between the bonds, the individual dipoles can either reinforce one another or cancel out each other.

- If the cancellation is complete, the resulting molecule will have no overall dipole and is said to be 'non-polar'.
- If the dipoles reinforce one another, the molecule will possess an overall dipole and is said to be 'polar'.

DIATOMIC MOLECULES

Hydrogen and chlorine are examples of diatomic molecules that are non-polar. The two atoms in each molecule are the same and so have the same electronegativity. The distribution of electron density of the bonding electrons in either molecule is totally symmetrical (see **Topic 3B.2**). The bond in each is therefore non-polar, making the molecules non-polar.

However, the bond in the hydrogen chloride molecule is polar because the electronegativity of chlorine (3.0) is greater than that of hydrogen (2.1).

$H^{\delta+} \rightarrow Cl^{\delta-}$

Since this is the only polar bond in the molecule, the molecule itself is polar.

The following symbol is used to represent a dipole: $\leftrightarrow$

The dipole in the hydrogen chloride molecule is shown as:

$H^{\delta+} — Cl^{\delta-}$

$\leftrightarrow$

POLYATOMIC MOLECULES

1. LINEAR MOLECULES

Example: carbon dioxide, CO_2

Both bonds in the carbon dioxide molecule are polar, but the dipoles cancel out one another.

$O^{\delta-}=C^{\delta+}=O^{\delta-}$

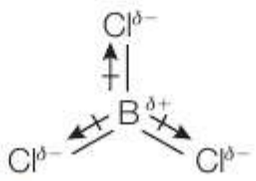
$\leftarrow\!\!+ \quad +\!\!\rightarrow$

▲ **fig A** Dipoles in carbon dioxide.

The carbon dioxide molecule is therefore non-polar.

2. TRIGONAL PLANAR MOLECULES

Example: boron chloride, BCl_3

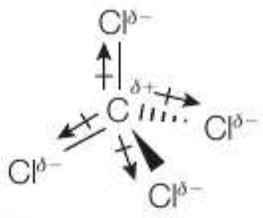


▲ **fig B** Dipoles in boron trichloride.

All three B—Cl bonds are polar, but because the molecule is symmetrical the dipoles cancel out one another. The molecule is non-polar.

3. TETRAHEDRAL MOLECULES

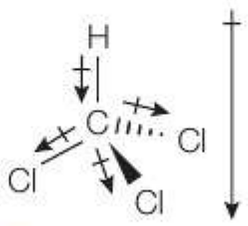
Example 1: tetrachloromethane, CCl_4



▲ **fig C** Dipoles in tetrachloromethane.

All four C—Cl bonds are polar, but because the molecule is symmetrical the dipoles cancel out one another. The molecule is non-polar.

Example 2: trichloromethane, $CHCl_3$



▲ **fig D** Dipoles in trichloromethane.

All four bonds are polar but, although the molecule is symmetrical, the dipoles reinforce one another and so the molecule is polar.

4. TRIGONAL PYRAMIDAL MOLECULES

Example: ammonia, NH_3

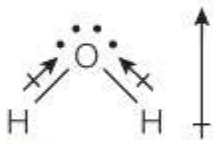
All three N—H bonds are polar and the dipoles reinforce one another. The molecule is polar.



▲ **fig E** Dipoles in ammonia.

5. V-SHAPED MOLECULES

Example: water, H_2O



▲ **fig F** Dipoles in water.

Both O—H bonds are polar and the dipoles reinforce one another. The molecule is polar.

ADDITIONAL READING

Dipole moments

The polarity of the molecule is measured by its **dipole moment**.

For a diatomic molecule such as hydrogen chloride, the dipole moment is defined as the difference in charge (i.e. the difference in magnitude between δ+ and δ–) multiplied by the distance of separation between the charges.

For a polyatomic molecule, it is more complicated because the polarities of each bond have to be taken into account, as well as any lone pairs on the central atom.

Table A gives the dipole moments of a number of molecules.

MOLECULE	DIPOLE MOMENT / D
H_2	0
Cl_2	0
HCl	1.05
CO_2	0
BCl_3	0
CCl_4	0
$CHCl_3$	1.02
NH_3	1.48
H_2O	1.84

table A Dipole moments of some molecules.

The unit of dipole moment is the Debye, symbol D. You do not need to understand this unit; just focus on the magnitude of the numbers. The larger the number, the more polar the molecule.

CHECKPOINT

1. A bond between two atoms in a molecule may possess a dipole.

(a) Explain how this dipole arises.

(b) Some bonds that you are likely to meet in organic chemistry are listed. Which of these bonds are likely to possess a dipole? In each case indicate which atom is δ+ and which is δ–.

C–Cl O–H C–C C–O C=C C–N N–H

2. State whether each of the following molecules are non-polar or polar. In each case, explain your reasoning.

(a) H_2S

(b) CH_4

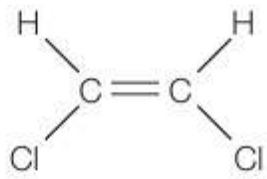
(c) SO_2

(d) SO_3

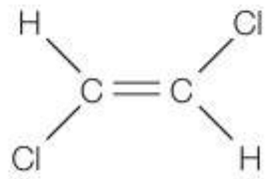
(e) $AlBr_3$

(f) PBr_3

3. There are two stereoisomers of dichloroethene. The structure and shape of each molecule is:



cis-dichloroethene



trans-dichloroethene

Suggest why the *cis* isomer is polar, while the *trans* isomer is non-polar.

SUBJECT VOCABULARY

dipole exists when two charges of equal magnitude but opposite signs are separated by a small distance

dipole moment the difference in magnitude between δ+ and δ– multiplied by the distance of separation between the charges